Lab 1 Rubric: Measurement and Uncertainty

Out of 100 points. This is the rubric on the LB1-M1 assignment in Brightspace. Lab 1 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read them before you write rather than after.

Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase "human error" earns no credit anywhere in the report, and writing it caps Discussion at Developing. Your references must include the source of your density table.

Criterion	Pts	Excellent	Good	Developing	Missing / Inadequate
Abstract	10	**10** Four sentences or fewer; states the theme (identifying four metals by density, imprecise vs. precise tools), the measured densities, and the % errors — nothing more.	**8** All three elements present but runs long or carries extra material, or a numerical result is stated incompletely.	**5** Missing one of the three required elements (theme, numerical densities, % errors), or reads like an introduction.	**0** Absent, or contains none of the required elements.
Introduction	10	**10** States $\rho = m/V$ and $V = \pi r^2 L$ with every variable described; gives background on how device precision sets uncertainty and significant digits; ends with the question the lab answers.	**8** Equations present but a variable undescribed, background thin, or the closing question implied rather than stated.	**5** An equation missing or undescribed; little or no background; no closing question.	**0** Absent, or no equations at all.
Experimental	10	**10** Reproducible blind: both tool sets; three trials each of length, diameter, and mass for all four rods, cycling through measurements; thread holder for the spring scale; how the Vernier scale was read; photo or labeled sketch.	**8** Reproducible in the main but missing one or two details (the cycling order, or the setup image).	**5** An outline only — a reader could not reproduce it; tool sets or trial structure unclear; no photo or sketch.	**0** Absent, or a sentence or two.
Results	25	**25** All four rods from both tool sets in numbered, captioned tables/figures, each introduced with "Table 1 shows…"; every measured value at the device's precision with matching uncertainty (caliper values carry more digits than ruler values); accepted densities stated with their source; one sample calculation each for an average, a volume, a density, and a % error.	**20** Data complete, but occasional sig-fig or uncertainty slips, a missing caption or opening sentence, or one sample calculation missing.	**13** Data incomplete (a rod or a tool set missing), sig figs and uncertainties largely ignored, or sample calculations mostly absent.	**0** Absent, or a raw spreadsheet dump with no presentation.
Discussion	20	**20** Opens with "We experimentally determined…"; identifies each rod's metal with its % error and a stated confidence; applies the 10% rule correctly; compares the precision of the two tool sets using the actual numbers.	**15** Identifications and % errors discussed, but the confidence claims or the tool-set comparison are thin, or the opening sentence is not in the given form.	**10** Generic accuracy talk; 10% rule misapplied or ignored; identifications asserted without numbers. Writing "human error" anywhere in the report caps this criterion at this level.	**0** Absent, or no engagement with % error at all.
Additional Questions	10	**10** All four comparison questions (most precise tool set, best answer, time, worth it) worked into prose with insight.	**8** All answered but superficially, or one question skipped.	**5** Several questions unanswered, or answers restated as a list.	**0** Absent.
Conclusion	5	**5** Restates densities and % errors, names each rod's metal definitively, and suggests specific measurement improvements.	**4** Results restated but improvements vague or identifications hedged.	**2** Numbers or the definitive statement missing.	**0** Absent.

Criterion	Pts	Excellent	Good	Developing	Missing / Inadequate
References	**5**	**5** Multiple sources in the numbered bracket format, including the density table actually used.	**4** Density source cited but it is the only reference, or the format is off.	**2** References present but the density-table source is missing.	**0** No references.
Format & Mechanics	**5**	**5** Written on the Lab Template; sentences and paragraphs throughout.	**4** Template used; one isolated list or formatting slip.	**2** Template ignored in places, or repeated bullet/numbered lists.	**0** Not on the template; lists throughout.
Total	**100**				

Overall: Excellent 90+ · Good 75+ · Developing 50+ · Missing / Inadequate below 50.

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